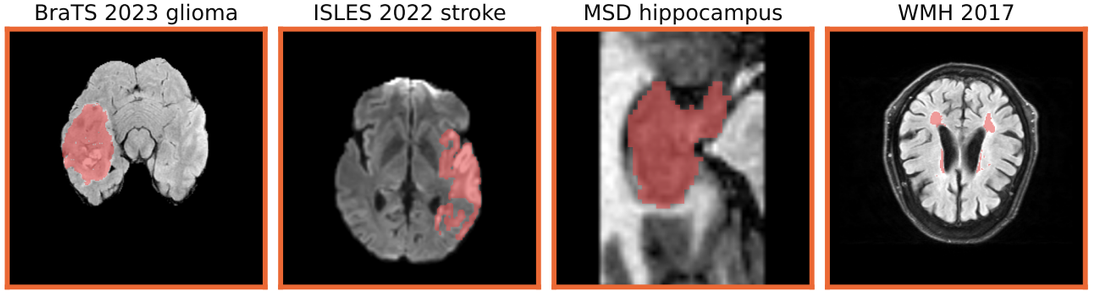
Which adaptation strategy for a new brain-MRI task?

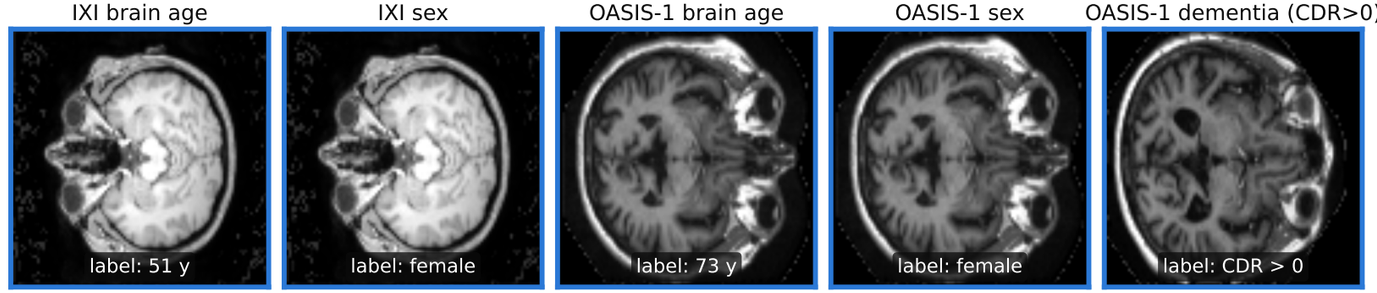
One rule decides it, and it is knowable before training: does the new task inherit a large task-specific head? Nine tasks, six cohorts, 32 labelled volumes each, frozen FOMO 3D U-Net, five seeds, 301 runs.

LARGE TASK HEAD · segmentation inherits an 8.45M-parameter decoder



BraTS 2023 glioma	ISLES 2022 stroke	MSD hippocampus	WMH 2017
3 ch · 1,251 vol.	DWI/ADC/FLAIR · 247	T1 · 260	FLAIR+T1 · 170 · 5 scanners

SMALL HEAD · a ~1K-parameter regressor or classifier over pooled features



IXI brain age	IXI sex	OASIS-1 brain age	OASIS-1 sex	OASIS-1 dementia (CDR>0)

→ Train the existing head (or LoRA, for the storage)

Train the decoder on the frozen encoder. LoRA + head stays within 0.015 Dice of it on all four tasks (BraTS -0.011, ISLES +0.002, hippocampus -0.011, WMH -0.015) while training 90× fewer parameters and storing 366 KB per task instead of 32 MB. Pick LoRA when storage matters, the decoder when it does not; full fine-tuning is more accurate still if the model will only ever serve one task (WMH +0.108).

→ LoRA is necessary, not merely economical

Use LoRA: it is the only frozen-backbone strategy that learns these tasks at all. Brain age MAE 11.0 y (IXI) and 11.9 y (OASIS) against constant baselines of 14.7 and 23.0; sex AUC 0.88 and 0.92. Linear probing and bias-only tuning return a constant output on every one of these tasks, even though their frozen features rank subjects far above chance: the signal is present, but a small head cannot move the representation to use it.

Only freezing prevents forgetting, in either order

Full fine-tuning: a second segmentation task erases the first whichever comes first — BraTS Dice 0.78 → 0.20 after ISLES, and ISLES 0.40 → 0.01 after BraTS. EWC forgets as much as plain fine-tuning (BWT -0.32 vs -0.33); LwF and replay forget more (-0.47, -0.40). LoRA and linear probing: BWT = 0 by construction.

Pretraining is a small, task-dependent effect

Same LoRA configuration on a randomly initialised backbone: identical on BraTS (+0.000 Dice) and IXI brain age, better without pretraining on ISLES (-0.030), and a modest gain elsewhere (WMH +0.024; sex +0.03 to +0.11 accuracy). Validate a checkpoint with a random-init control; do not assume transfer.

A protocol that cannot flatter a method

Subject-grouped train / val / test splits, whole-volume sliding-window inference, per-case Dice and HD95, five seeds, 301 runs, paired per-case tests with Holm correction. The pipeline we started with scored random crops on the selection split and inverted the ranking of methods. Code, splits and predictions released.